

# AI Effects on the Labor Market:

## Is AI Disrupting the Future of Work?

# How to read this presentation

This Deep Dive explains how Artificial Intelligence (AI) is reshaping the labor market from first principles.

The sections are designed to be read in order and assume no prior background knowledge.

Our goal is not to predict the future with certainty, but to give you a clear map of the mechanisms at play: how automation, augmentation, and new-task creation interact to shape jobs, wages, and productivity.

For nearly every slide in this deck, there are academic papers and policy reports that explore these topics in far greater depth.

Our goal isn't to overwhelm you with data, but to make complex dynamics legible. Omissions are intentional.

/ Learn With Me

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# The Mechanics That Drive AI's Impact on Labor



# Experts Agree AI Will Transform Jobs, but Disagree on How.



"I think most companies are going to be doing more or less the same things. A few occupations will be impacted, but we're still going to have journalists, we're still going to have financial analysts, we're still going to have HR employees."

- Daron Acemoglu, Nobel Laureate in Economics



"There's tremendous opportunity here. I do not think that AI means the death of all human expertise... it means amplifying in many realms and supplementing it for people who have judgment but who are not at the frontier."

- David Autor, Professor in the MIT Department of Economics, Co-Director of Labor Studies NBER



"AI will be a massive productivity boost for existing job roles and it will create many new job roles... there will be some job loss, but I think it may not be as bad as people are worried about right now."

- Andrew Ng, Adjunct Professor at Stanford University, Founder & Lead for the Google Brain Project



"You won't lose your job to AI, but you will lose your job to someone who uses AI."

- Jensen Huang, President and CEO of Nvidia



"AI could wipe out half of all entry-level white-collar jobs."

- Dario Amodei, Co-founder and CEO of Anthropic



# What Is a Job?

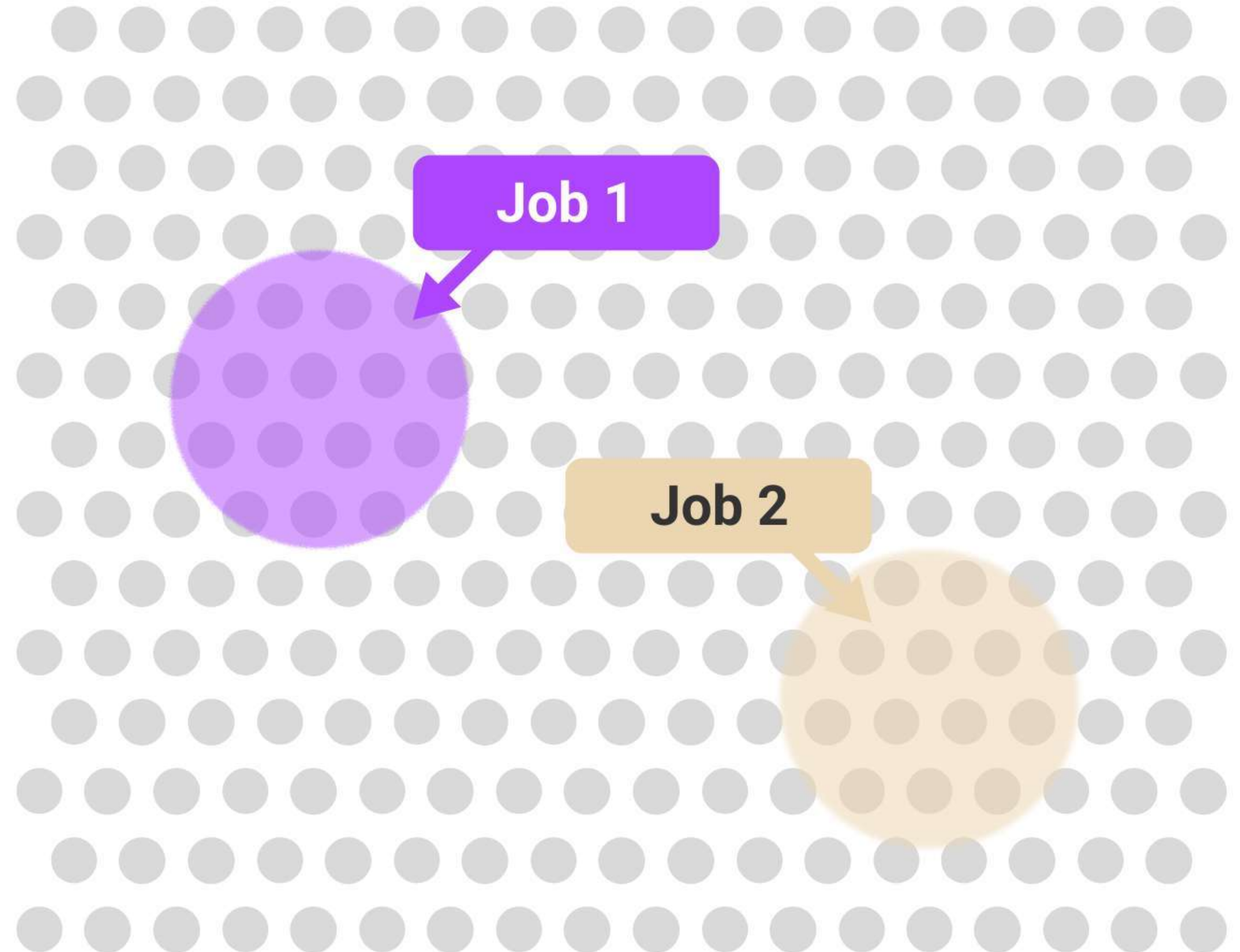
At its core, a job is a bundle of tasks.

A doctor diagnoses symptoms, orders tests, interprets results, and coordinates with specialists.

A software engineer writes code, fixes bugs, reviews pull requests, and explains trade-offs to stakeholders.

A lawyer drafts contracts, researches case law, argues motions, negotiates settlements, and advises clients.

## Universe of Tasks



# A Three-Layer Framework for Thinking About AI and Jobs

When a big technology arrives, like AI, it almost never wipes out whole jobs overnight.

It affects *tasks* first.

Some tasks get automated, some get supercharged, and some create new kinds of work.

To make sense of this, it helps to look at AI's impact through three layers:

(1) Exposure

(2) Adoption

(3) Labor market response



# Layer 1: Exposure

This layer answers: **Given current and foreseeable capabilities, which tasks in a job are susceptible to automation, which are augmentable, and which are resistant?**

To answer, we need two components:

- 1) **The task composition of each occupation.** This allows analysis of the actual bundle of tasks in a computer programmer's or a nurse's job.
- 2) **The pattern of AI development,** for example today's models excel at text and pattern recognition but are weak at dexterous manipulation or social nuance.

Researchers combine these two components into an “exposure index,” which estimates how much of a job current AI could realistically assist or perform.

Crucially, **exposure isn't a forecast of layoff.** It just tells you where the technical potential lies, not whether firms or organizations will deploy it. That brings us to Layer 2.



## Layer 2: Adoption

This layer answers: **How quickly and in what form will organizations deploy AI?**

Even if a task is technically automatable, adoption can lag because of cultural and institutional factors like legal constraints, union rules, customer backlash, or lack of complementary investments.

The degree of adoption shapes Layer 3.

## Layer 3: Labor Market Response

This layer answers: **Given the pattern of adoption, what is the net effect on employment, skill demand, wages and the creation of new jobs?**

Economists talk about three response types:

- 1) Augmentation:** The labor market responds by keeping humans in their current jobs but making them more productive by “augmenting” them with AI tools.
- 2) Displacement:** The labor market responds by displacing human workers and replacing them with machines.
- 3) Reinstatement:** The labor market responds by creating new tasks and expanding human roles that complement technology. This produces new jobs such as AI engineering and other times existing occupations broaden as productivity gains lower costs and raise demand.

Here, labor market response reflect the collective decisions of firms, workers, and policymakers.

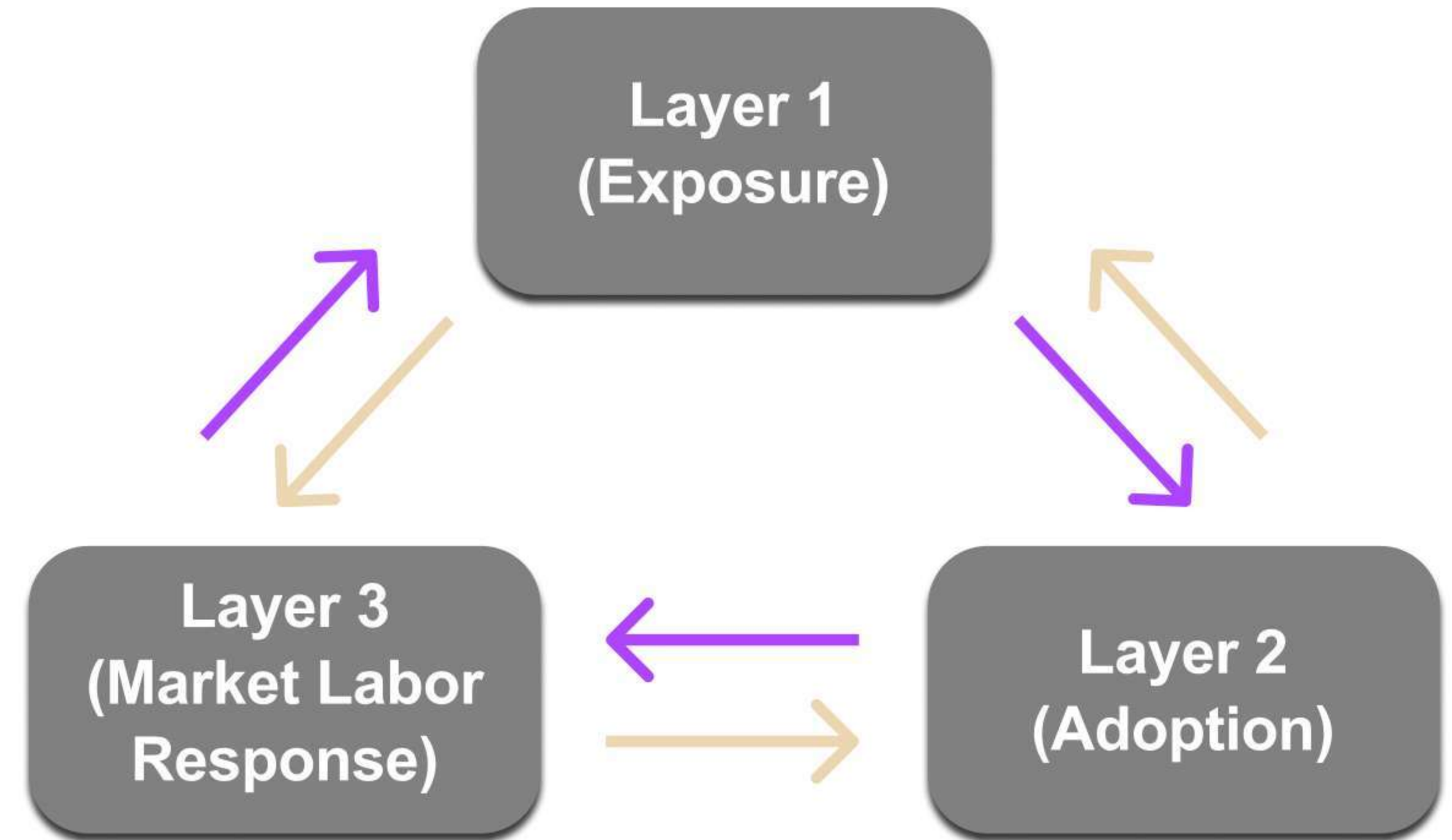


# Feedback Loops

The three layers are not sequential. They constantly feed back into each other in multi-directional ways, as we'll see later.

This means that the same technology can yield different outcomes depending on how these factors interact. This also explains why the discussion on AI's impact on the labor market is so contentious.

If the impact of AI depends on feedback loops between technology, firms, and workers, then the first step is to trace where those loops begin.





# The Scale of Potential Impact

The U.S. economy is \$31 trillion. Almost 60% of that is labor costs. So even if AI only affects a small percentage of jobs, we're talking about trillions of dollars shifting around the economy.

Globally, analysts estimate that as many as 300 million full-time jobs, roughly 27% of positions in OECD countries, will be transformed through AI automation.

However, history shows that technological revolutions rarely lead to job loss overnight.

The real challenge lies in the pace and distribution of change, who adapts fastest, who benefits most, and how policy and institutions respond.

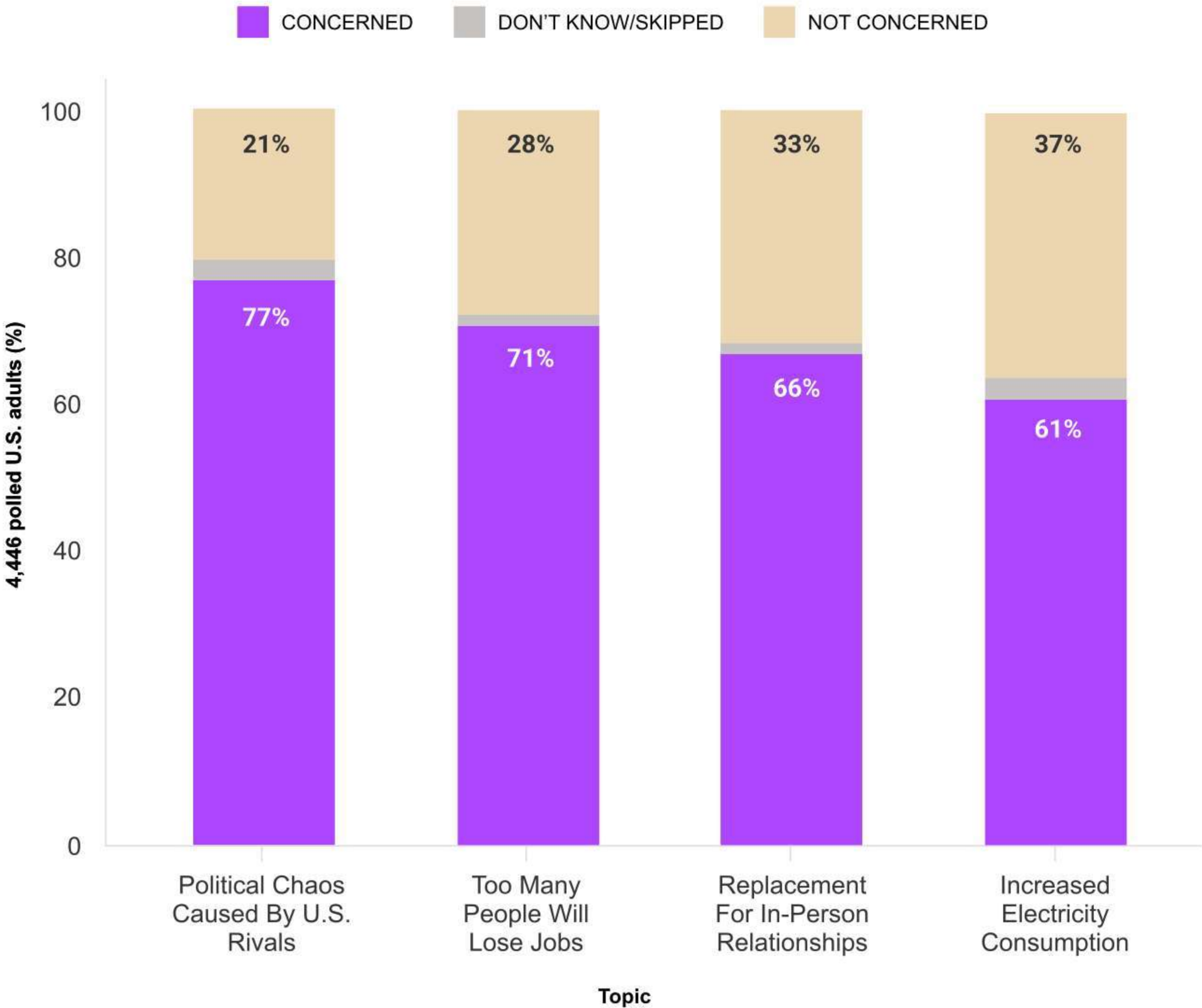


# Early Concerns

71% of respondents in an August 2025 poll said they were concerned AI will be "putting too many people out of work permanently."

This captures a prevalent concern, but the actual impact is more nuanced and includes upside effects.

American Concerns About Artificial Intelligence

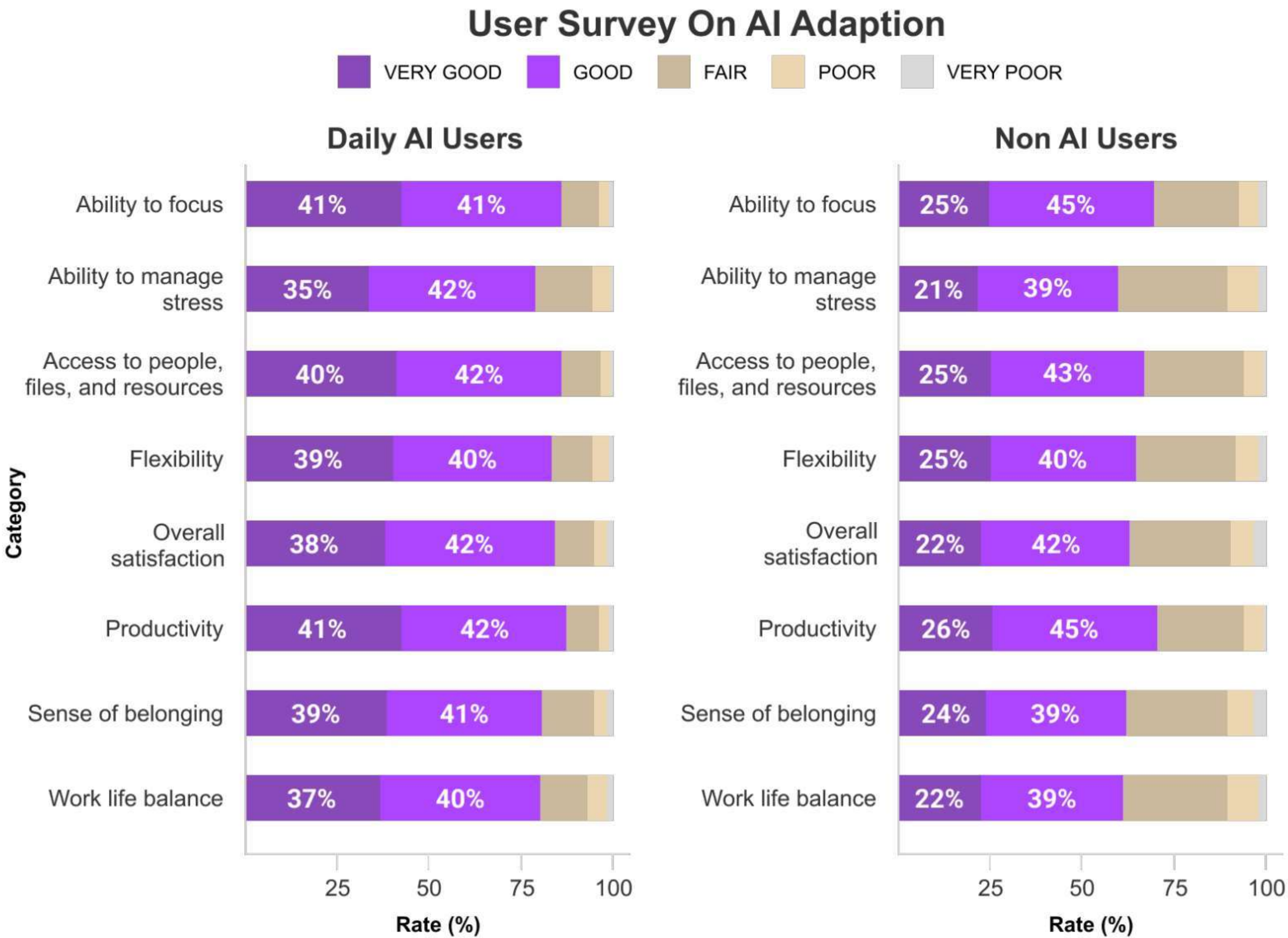




# Improvement with Usage

The latest Workforce Index from Slack shows that daily AI users consistently outperform non users across key work experience metrics, including productivity, focus, stress management, and job satisfaction.

Across nearly every dimension, those who actively use AI report meaningfully better outcomes than peers who do not.





# This Deep Dive

## Our Approach:

This deck focuses on the past, present and the next 6-24 months rather than artificial general intelligence scenarios or robotics.

A future deep dive is coming for humanoid robotics, so keep an eye out for that.

We'll analyze AI's impact on work by breaking jobs down into tasks and by looking at three interconnected layers that determine real-world outcomes. We'll look at how exposure is measured and why exposure alone doesn't tell the full story.

# This Deep Dive

## Questions this Deep Dive will answer:

1. How is AI affecting the labor market today?
2. What is the scale and scope of AI adoption?
3. What to watch out for and who will benefit from AI productivity gains?



# Layer 1: Exposure

# The Exposure Index

Think of an “exposure index” as a measure from 0 to 100 that estimates to what extent the tasks in your job can be plausibly assisted or performed by current or near-term AI.

The higher the score, the more your job might be impacted.



# How Do We Measure Exposure?

There are many different approaches to construct exposure indices, each giving a different ranking of which jobs are most open to change.

One approach is “Task to Ability Mapping.” It identifies the tasks that make up a job, estimates which of those tasks current AI can handle, and weights them by their importance in the occupation. The resulting measure is “AI Occupational Exposure” (AIOE).

This method produces tables that rank jobs and industries from most exposed, for example genetic counsellors, to least exposed, for example dancers.

Occupations with the Highest and Lowest AIOE Measures

| Rank | Highest scoring                                                | Lowest scoring                                                               |
|------|----------------------------------------------------------------|------------------------------------------------------------------------------|
| 1    | Genetic counselors                                             | Dancers                                                                      |
| 2    | Financial examiners                                            | Fitness trainers and aerobics instructors                                    |
| 3    | Actuaries                                                      | Helpers - painters, paperhangers, plasterers, and stucco masons              |
| 4    | Purchasing agents, except wholesale, retail, and farm products | Reinforcing iron and rebar workers                                           |
| 5    | Budget analysts                                                | Pressers, textile, garment, and related materials                            |
| 6    | Judges, magistrate judges, and magistrates                     | Helpers - Brickmasons, Blockmasons, stonemasons, and tile and marble setters |
| 7    | Procurement clerks                                             | Dining room and cafeteria attendants and bartender, helpers, clerks          |
| 8    | Accountants and auditors                                       | Fence erectors                                                               |
| 9    | Mathematicians                                                 | Helpers - roofers                                                            |
| 10   | Judicial law clerks                                            | Slaughterers and meat packers                                                |
| 11   | Education administrators, postsecondary                        | Landscaping and Groundskeeping workers                                       |
| 12   | Clinical, counseling, and school psychologists                 | Athletes and sports competitors                                              |
| 13   | Financial managers                                             | Fallers                                                                      |
| 14   | Compensation, benefits, and job analysis specialists           | Structural iron and steel workers                                            |
| 15   | Credit authorizers, checkers, and clerks                       | Cement masons and concrete finishers                                         |
| 16   | History teachers, postsecondary                                | Terrazzo workers and finishers                                               |
| 17   | Geographers                                                    | Rock splitters, quarry                                                       |
| 18   | Epidemiologists                                                | Plasterers and stucco masons                                                 |
| 19   | Management analysts                                            | Brickmasons and Blockmasons                                                  |
| 20   | Arbitrators, mediators, and conciliators                       | Roofers                                                                      |



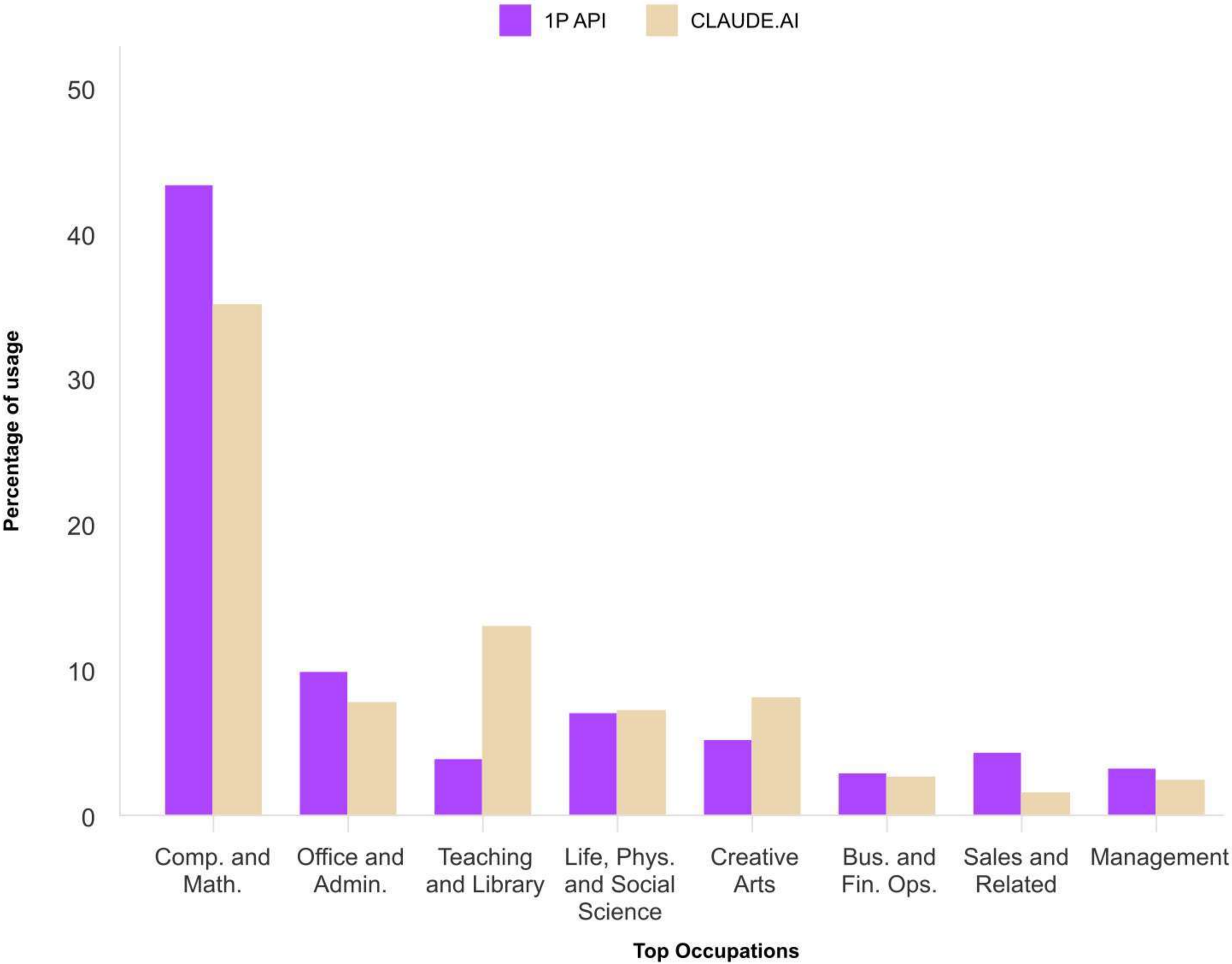
# How Do We Measure Exposure?

Another approach is to analyze what workers actually use AI for today by studying millions of AI conversations and mapping them to specific job tasks, resulting in indices like Anthropic’s Economic Index.

Example: Anthropic analyzes 1+ million Claude conversations to see which occupations are actively using AI for which tasks right now. Currently, 44% of AI usage is for computer and mathematical analysis.

OpenAI also launched their own performance measure with GDPval on September 2025.

Usage Shares Across Top Occupational Categories



Source: Anthropic Economic Index (2025)



# Which Jobs are Most Exposed?

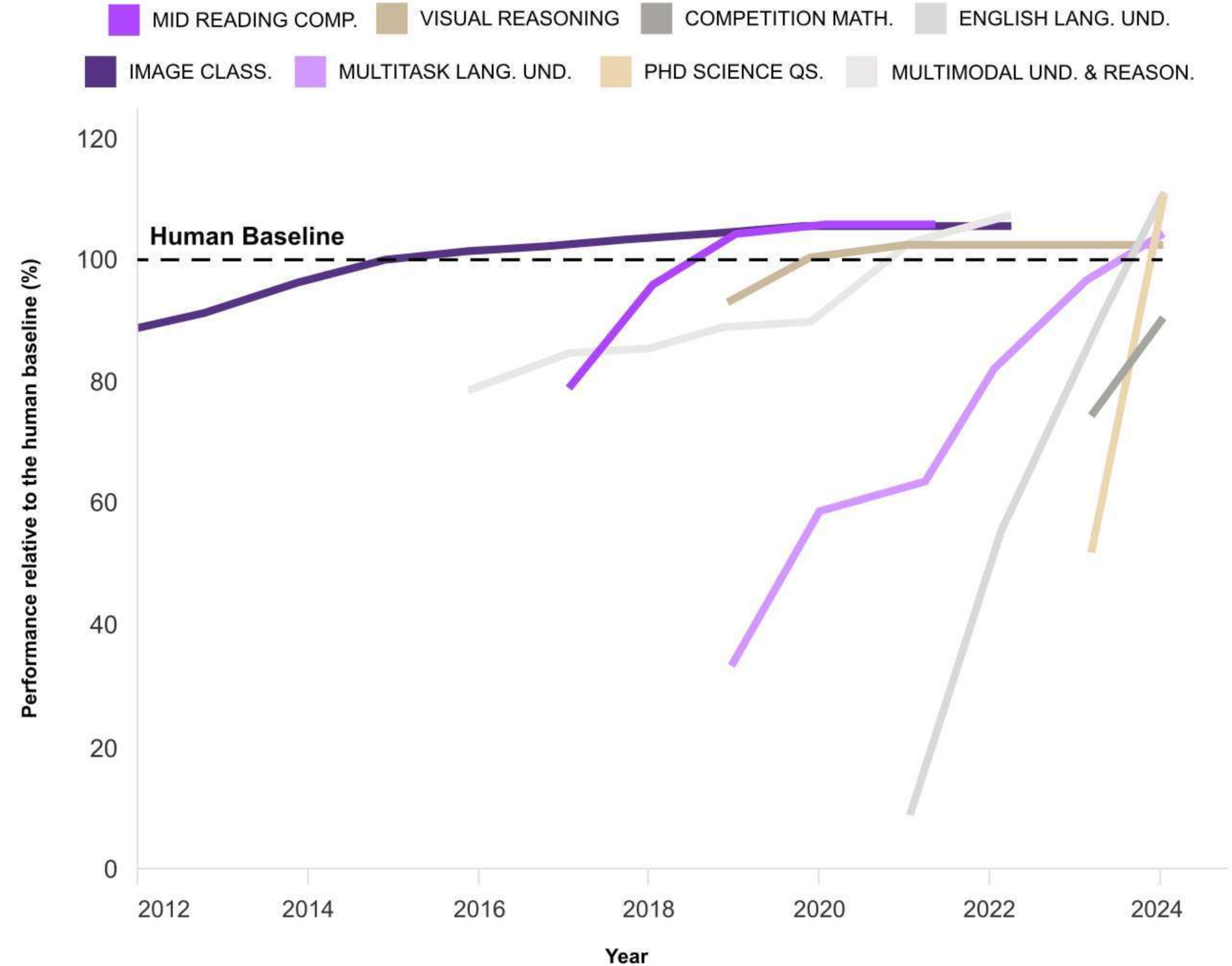
Although the technical details of the various exposure indices differ, recent studies consistently point to several patterns:

1) Text-heavy, information-processing, and communication tasks are the most exposed.

Legal research/drafting, writing/editing, customer support, market research/analytics, education content prep, finance/insurance analysis, and programming documentation/QA consistently score high on the exposure index.

Because LLMs are trained on vast text corpora and then refined through reinforcement learning from human feedback, they excel at tasks with abundant written examples and clear preference signals.

AI Technical Performance vs Human Performance (2012-2024)



# Routine Roles

2) Clerical and back-office roles are one of the most automatable within the exposed set.

Routine document processing, booking, transcription, and standardized communications are repeatedly flagged as highly affected and plausibly automatable.





*“When looking for jobs that will change a lot due to AI on shorter time scales, I'd look in other places - jobs that look like repetition of one rote task, each task being relatively independent, closed (not requiring too much context), short (in time), forgiving (the cost of mistake is low), and of course automatable given current (and digital) capability.”*

Andrej Karpathy  
Former Director of AI @ Tesla and founding team @ OpenAI  
2025



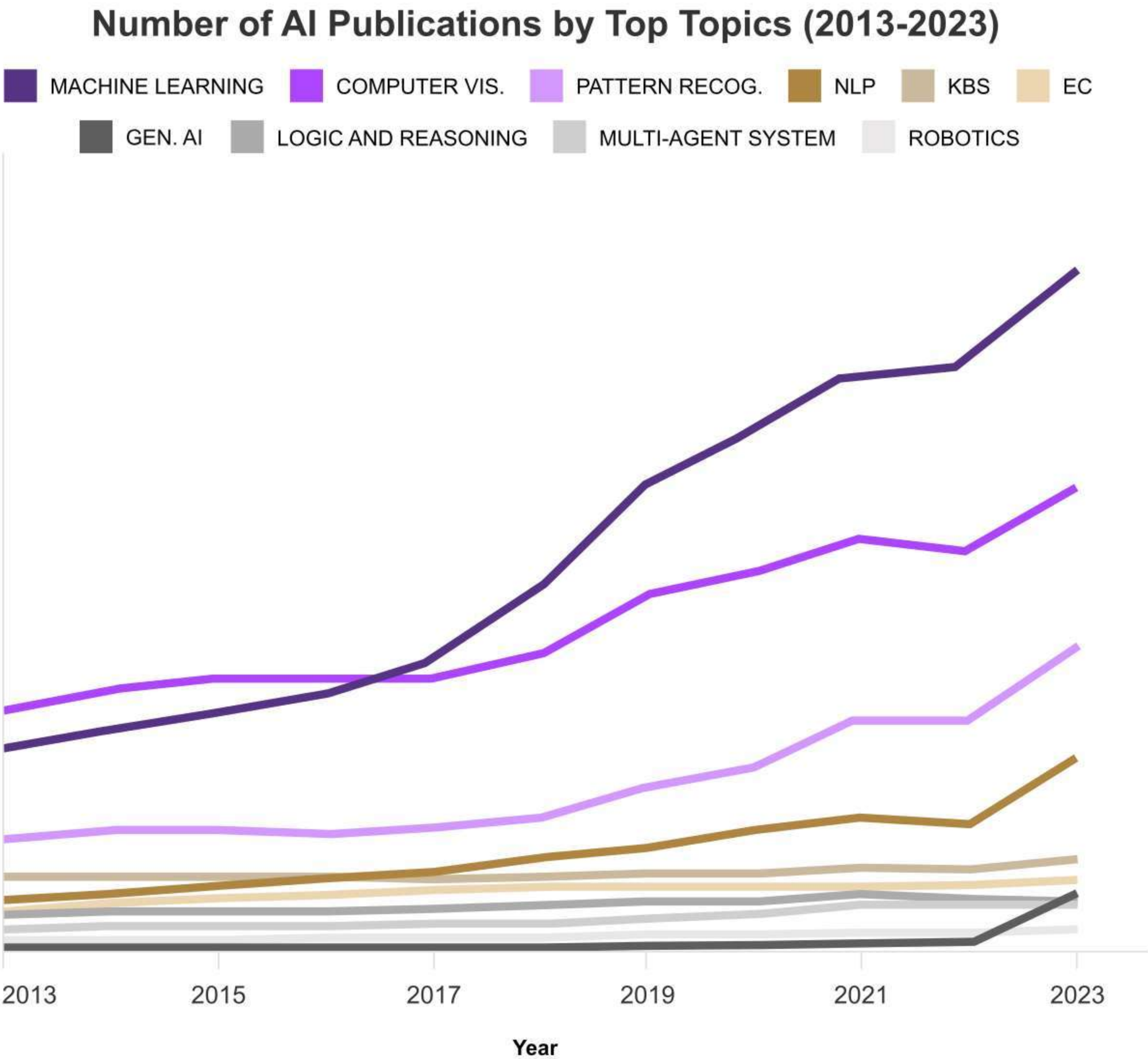
# Least Exposed

3) Embodied, on-site, and interpersonal-dexterity tasks are currently least exposed.

Electricians, roofers, line workers, nursing assistants, and healthcare aides. Roles requiring physical manipulation, presence, or rich social interaction score low in current exposure indices.

This highlights the fact that most AI is currently disembodied and that the boom in AI is happening much faster than the evolution of practical robots (though that might change soon).

This is also reflected by the difference in growth and levels of publications on ML vs robotics in recent years.





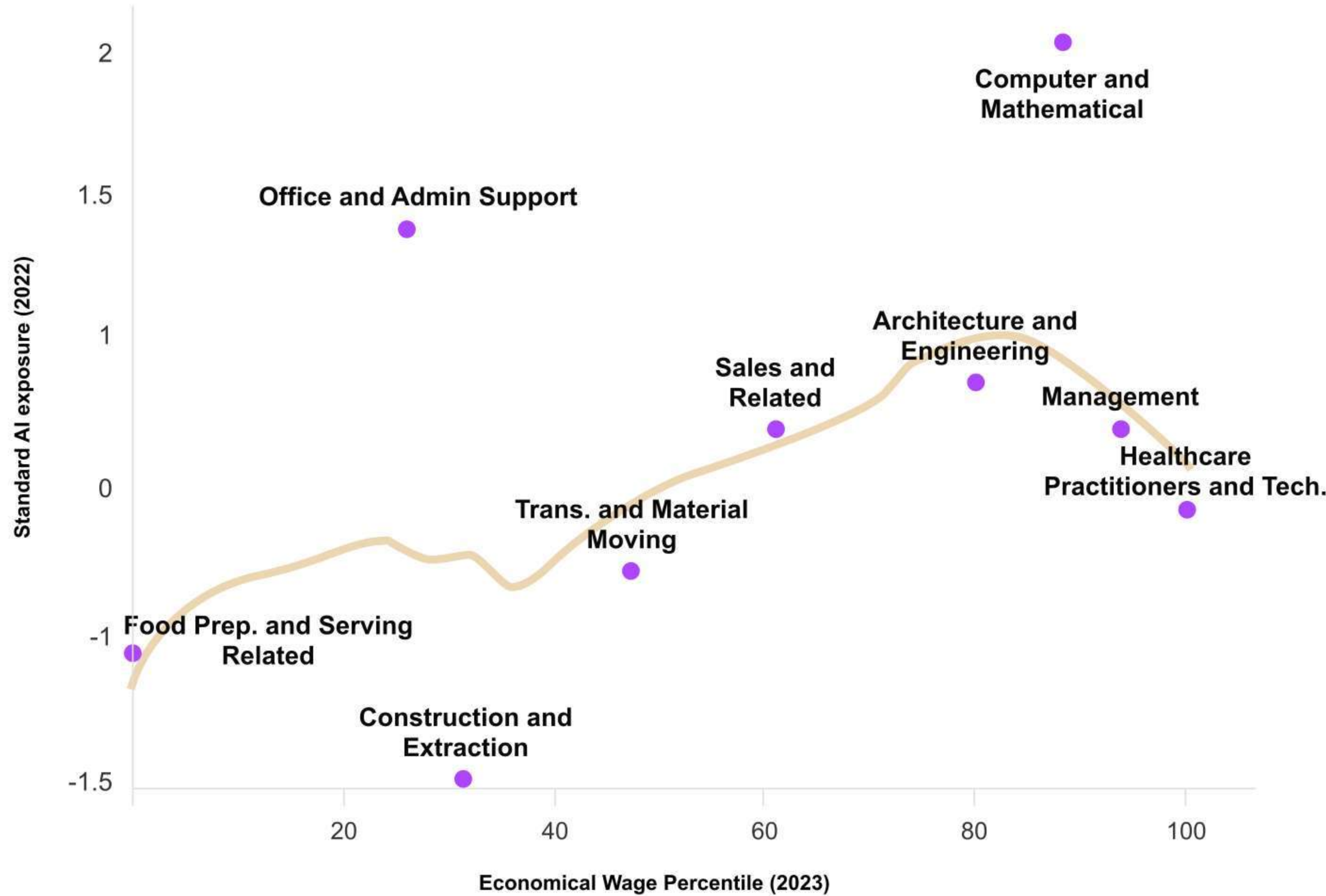
# Exposure Correlation

## 4) Exposure is positively related to wages and education

Higher paid and higher educated occupations are more exposed to LLM era AI because their work is language and symbol dense.

In contrast to earlier tech waves that automated routine or physical tasks, Gen AI reaches cognitive non-routine work once assumed to be safe from automation.

Generative AI Exposure by Occupational Group



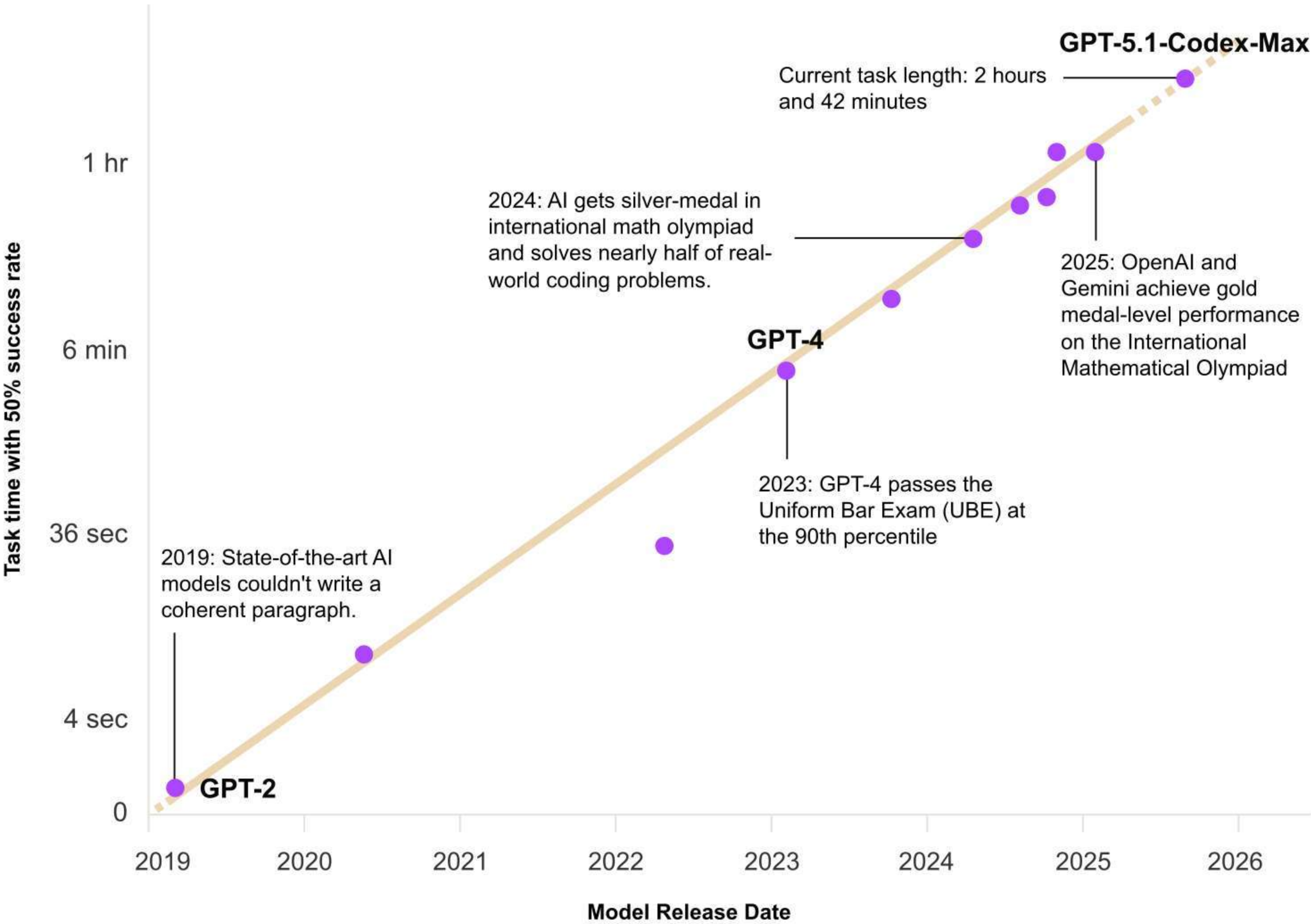


# A Moving Measure

Although existing exposure indices paint a similar picture in terms of who's exposed, we should treat current tables and graphs as constantly evolving as the tech continues to advance.

Many analysts are trying to model the advancement of AI through measures of task time completion with a success rate over 50%.

AI Agents' Capacity for Autonomous Task Completion (Log Scale)





# The New Moore's Law

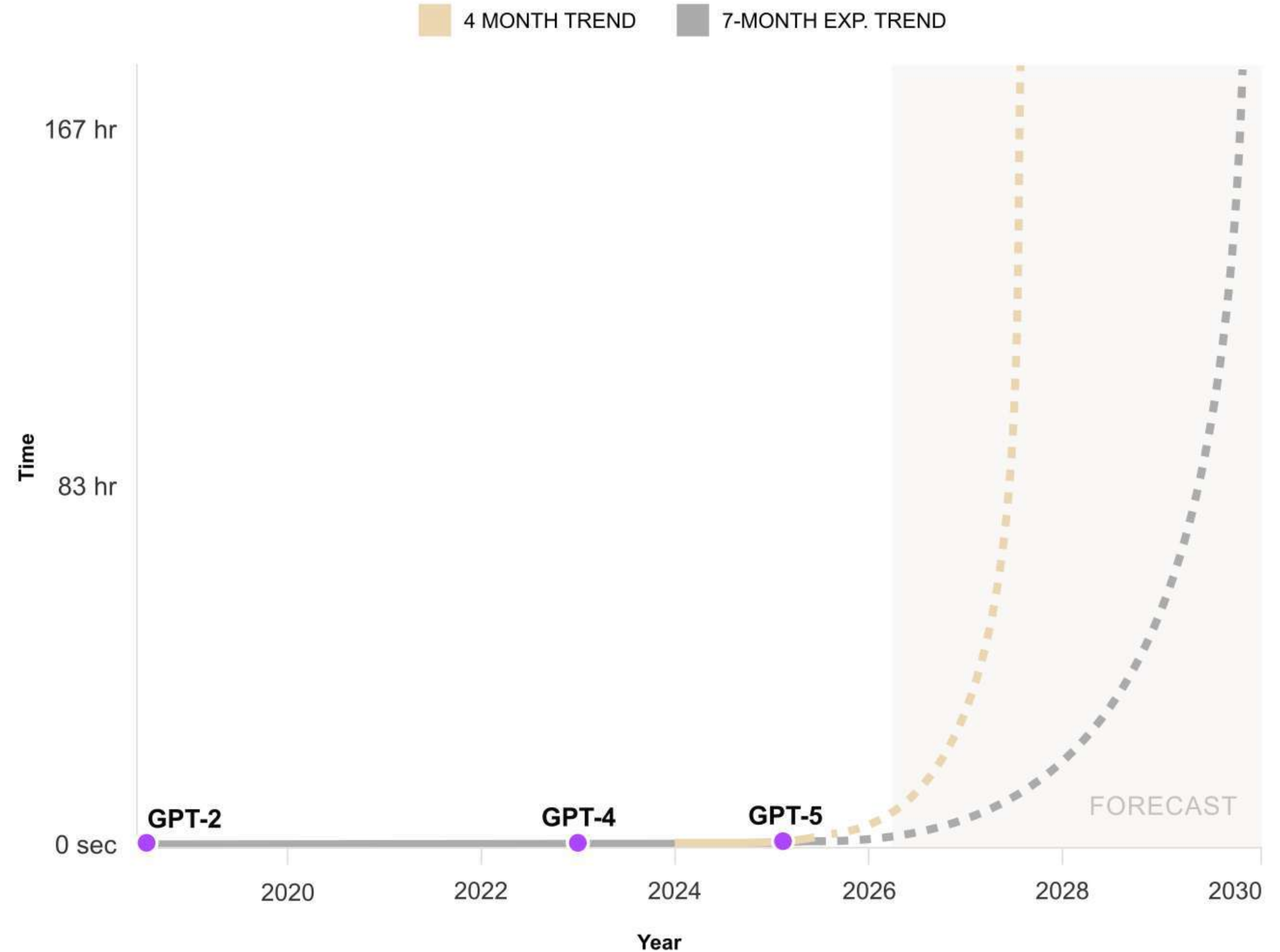
This speed of progress in the past 5 years has led researchers to formulate the “New Moore’s Law”, which says the length of tasks AI can perform doubles every ~7 months.

Assuming an exponential trend in capability growth, this trend predicts that...

- By 2026, AI can complete a 2-hour task
- By 2027, AI can complete 1 work day (8 hours)
- By 2028, AI can complete 1 work week (40 hours)
- By 2029, AI can complete 1 work month (167 hours)

Recently, there has been some evidence that the trend is ramping up to doubling every ~4 months.

## Projected AI Advancement in Coding Task Autonomy





# Experts Have Been Wrong In The Past

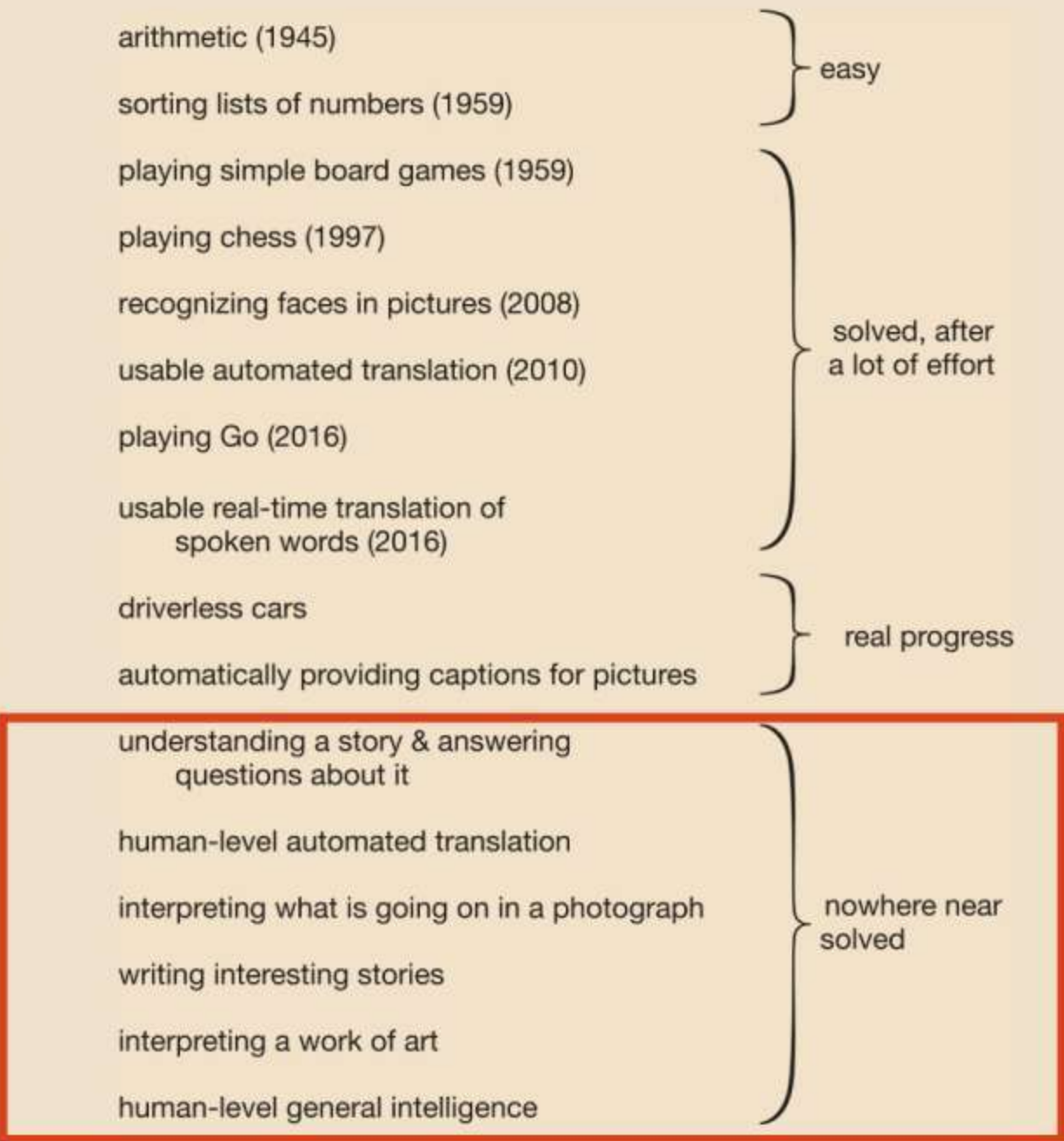
While these exponential projections are tidy in theory, experts have been wrong in both directions, underestimating and overestimating the speed of progress.

An example of underestimation: In 2021 Michael Wooldridge, a leading Oxford AI researcher, published a book listing tasks he judged nowhere near solved: understanding a story, human level automated translation, interpreting a photograph, and writing interesting stories.

These are things that your favorite LLM can easily do today.



—but others are not. To understand why AI is hard—why progress has proved so elusive over the years—it helps to look at problems that are easy to encode in this way and problems that are not, and to see why this is the case. **Figure 1** displays some tasks that we might want computers to do and shows how difficult it has proved to get computers to achieve them.



**Figure 1:** Some tasks that we want computers to be able to do ranked in order of difficulty. Years in parentheses indicate approximately when the problem was solved. At present, we have no idea about how to get computers to do the tasks at the bottom of the list.





## Experts Have Been Wrong In The Past

*“I think if you work as a radiologist you're like the coyote that's already over the edge of the cliff, but hasn't yet looked down, so it doesn't realize there's no ground underneath him. People should stop training Radiologists now. It's just completely obvious that within 5 years deep learning is going to do better than Radiologists...”*

*He was right about the models outperforming radiologists on benchmarks, but wrong about the demand.*

*The field of Radiology continues to grow and stay strong, a phenomenon we'll explore later.*

Geoff Hinton “Godfather of AI”  
Machine Learning and Market for Intelligence Conference  
2016



# Exposure Recap

Exposure is an estimated measure of how much of a job's daily work could be performed by AI.

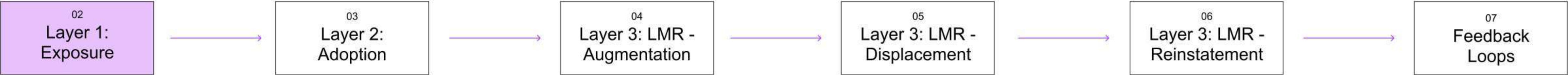
A job that is highly exposed can also be the one most likely to increase in growth as the role adapts to the rise in productivity.

Across studies, a pattern emerges: high-paying, highly educated jobs that rely heavily on language and symbols, from law and finance to marketing and software, score higher on AI exposure.

In contrast, low-wage or physically anchored jobs remain least exposed.

We've seen that AI's capabilities have raced ahead of expectations, but capabilities alone don't reshape economies.

We need to consider how AI is adopted through firms and workflows, which we'll examine in the next section.





# Layer 2: Adoption

# AI Adoption and Diffusion

Exposure tells you what AI could do. Adoption and diffusion tell you what AI is actually doing in the world.

Adoption refers to an individual or firm's process of deciding whether to use a new innovation.

Diffusion is the broader societal process of how that innovation spreads over time through a population.

For the purposes of these slides, we'll simplify both under the single category of adoption.

Exposure doesn't always equal adoption. A job can be highly "exposed" in theory but untouched in practice.

For example, a firm may ban their workers from using public LLMs for data confidentiality reasons or hasn't reorganized their workflows to use the new tool efficiently.



*“Even if a new AI-based product is instantly released online for anyone to use for free, it takes time for people to change their workflows and habits to take advantage of the benefits of the new product and to learn to avoid the risks.”*



Arvind Narayanan & Sayash Kapoor  
Computer Scientists at Princeton University  
AI as normal technology (2025)



# AI as a General-Purpose Technology

AI sits in the family of general-purpose technologies (GPT\*). These are innovations so broadly applicable that they transform not just one sector but the entire economy.

The steam engine, electricity, and the personal computer are classic examples of GPTs.

GPTs are different from narrow innovations. A new fertilizer might revolutionize agriculture but won't touch finance or healthcare.



# Historical Adoption Rates for GPTs

The key thing to know about GPTs in the context of adoption is that they rarely produce instant productivity gains.

For example, electricity took decades to raise factory output because many factories couldn't get wired up immediately.

Building the power grid, transmission lines, motors, and wiring was expensive.

So even when electricity was available, factories often replaced steam engines with electric motors as opposed to fully electrifying operations.



# The Productivity J-Curve

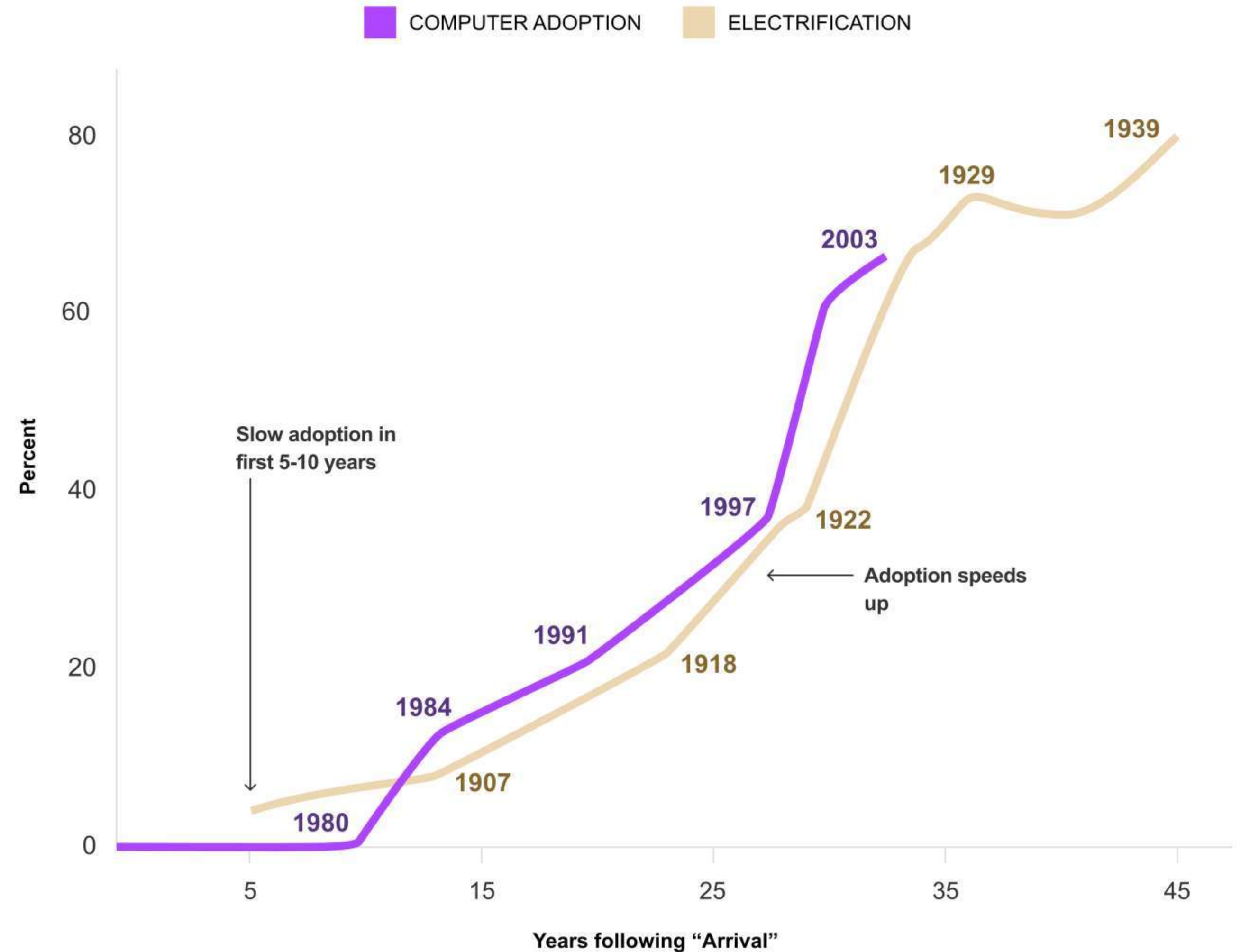
Another example is the personal computer (PCs).

Although PCs became available to firms in the early 1980s and could theoretically make some workers more productive, availability wasn't enough to lead to immediate productivity gains.

Firms needed training and complementary tools to realize the theorized productivity gains.

The broad productivity gains from personal computers became visible in U.S. statistics only around the mid-1990s, as complementary software, networking, and new work practices matured.

The Pace of Technological Adoption Since GPT Arrival







# The Productivity Paradox

*Economists noted the productivity paradox through the late 1980s and early 1990s.*

*“You can see the computer age everywhere but in the productivity statistics.”*

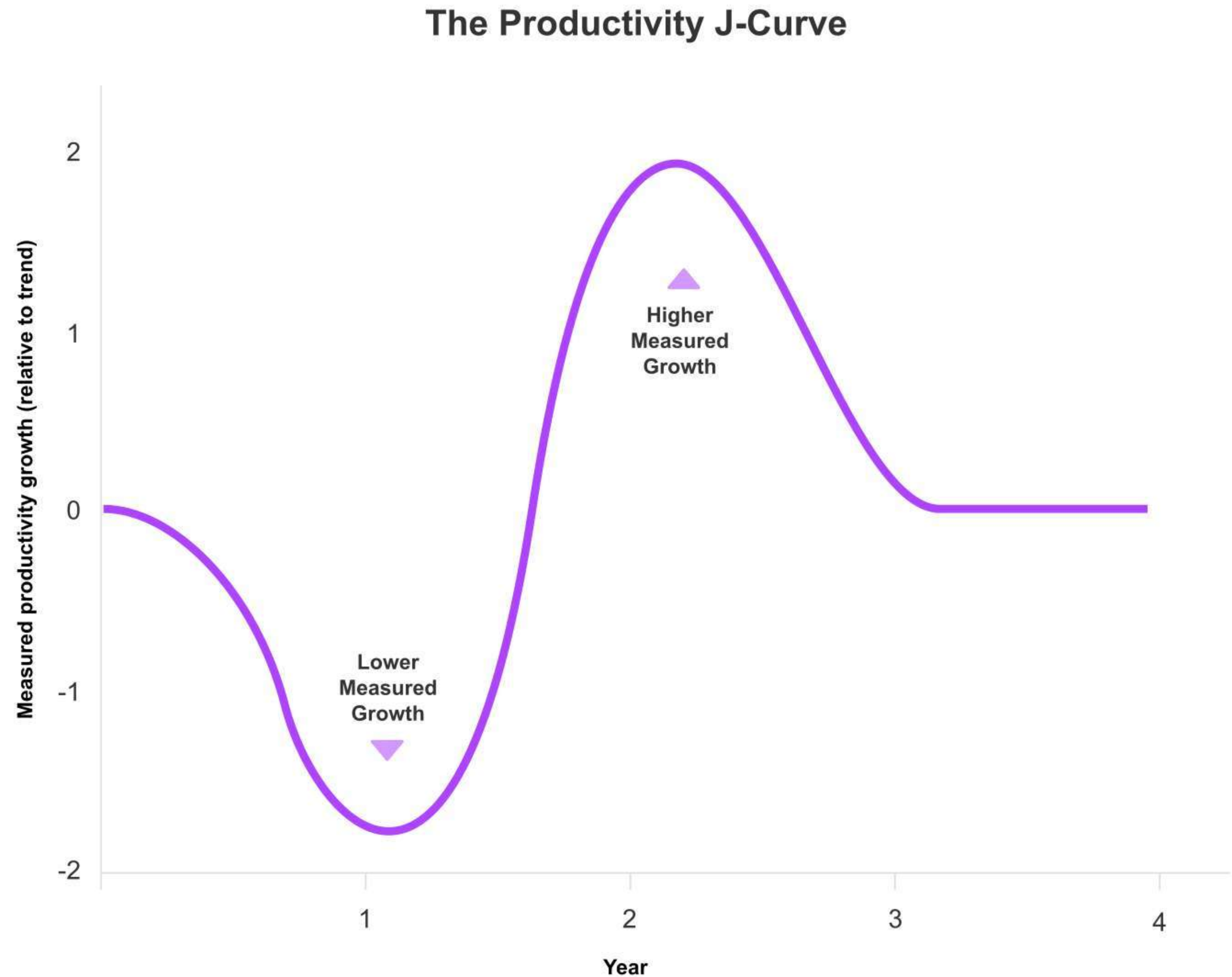
Robert Solow  
Nobel Laureate in Economics  
1989

# The Productivity J-Curve

In early years after implementation, productivity often grows slowly or even dips due to learning costs, setup, and process re-engineering.

Historical examples of GPT adoption results in “The Productivity J-Curve”.

When we plot the relationship between productivity growth rate on the Y-axis, against time since the technology is introduced on the X-axis, we see a J-shape due to productivity appearing to dip during the investment phase, overshooting as intangible gains are realized, and then normalizing at a higher level of output.





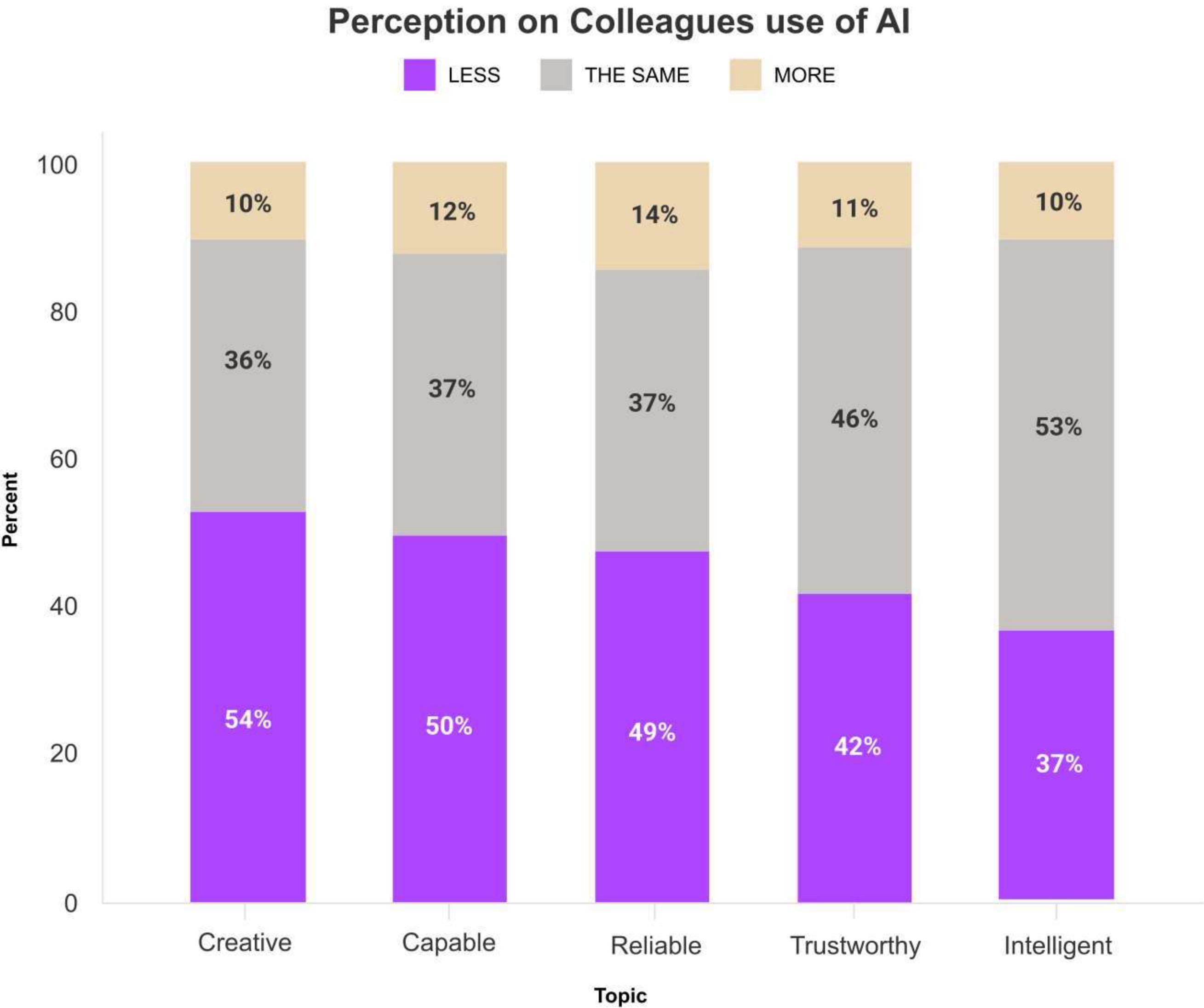
# The Productivity J-Curve

A 2025 Harvard Business Review study found 40% of knowledge workers reported cleaning up such “workslop,” spending nearly 2 hours per week fixing low-quality outputs.

This suggests that early adopters might initially see higher activity, but not necessarily higher output value.

When firms invest in infrastructure, train employees, and redesign workflows, costs tend to fall as benefits accumulate.

Eventually productivity accelerates, rising sharply as adoption becomes widespread.



# Pause and Performance

Another factor is the potential constant switching cost created by rapid AI launches, which can force repeated workflow resets.

But if models pause or plateau at a level like ChatGPT 5.2, Gemini 3, or Opus 4.5, productivity gains can still emerge from:

- Firms and workers getting better at using the AI tools they already have.
- Integrating AI into workflows.
- Automating tasks that are ripe for automation.
- Shifting worker effort from low-value tasks to higher-value ones.
- Scaling up adoption from early adopters to the broader economy.



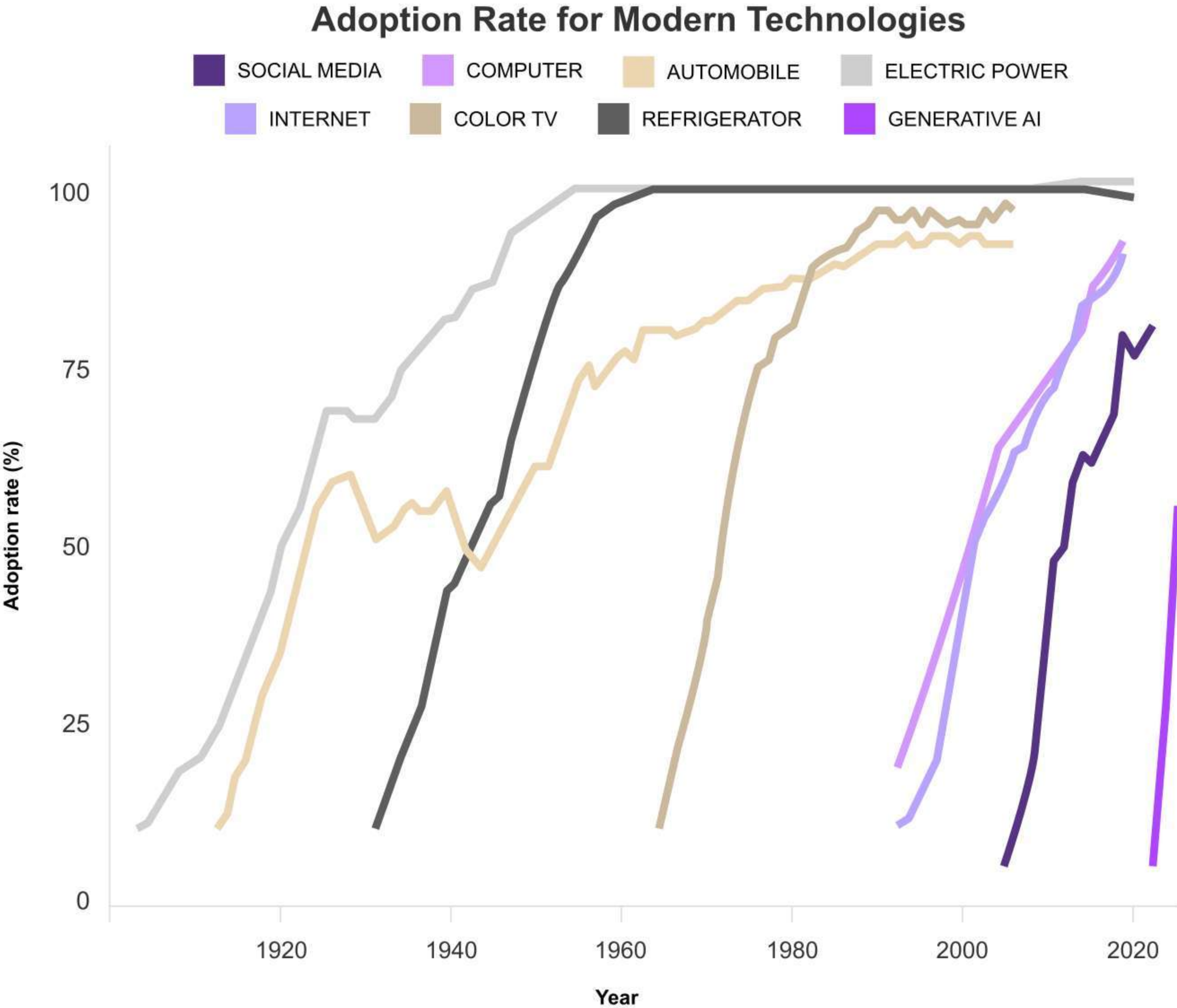
# Adoption Rates For Modern Technologies

Adoption rates for modern technologies, such as social media, are faster than for older technologies like the refrigerator.

It took 50 years for US households to go from 10% adoption of electricity to 99%.

In contrast, it took just 15 years for social media to go from 5% adoption to 79%.

Within three years of ChatGPT's 2022 launch, the adoption rate of generative AI has surpassed 50%.





# What We Know About AI Adoption So Far

**Worker-Level** (U.S. population age 18-64 from The Real-Time Population Survey)

The survey found that in August 2024, 44.6% of adults used generative AI overall.

Within a year, the adoption of this new technology increased by 10 percentage points to 54.6%.

